

Dr. Tarun Kumar Kulshrestha

AI/ML Research Scientist | Deep Learning Engineer | Generative AI & Computer Vision

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PROFESSIONAL SUMMARY

Deep Learning and Generative AI researcher with 10+ years across applied research, engineering, and academic technology enablement. Ph.D. from IIT Roorkee and 5+ years at The Hong Kong Polytechnic University/UBDA. Experienced in converting research and proposal requirements into executable ML/DL workflows, guiding GPU/HPC experimentation, and publishing at CVPR, AAAI, WACV, IEEE TIP, and IEEE TMC. Expertise spans LLMs, RAG, GANs, computer vision, data-efficient learning, model optimization, and practical AI deployment.

CORE TECHNICAL SKILLS

AI / Deep Learning	Generative AI, LLMs/SLMs, RAG, LangChain, GANs, CNNs, RNNs, transfer learning, image-to-image translation, multimodal AI, few-shot/data-efficient learning, pruning, quantization, efficient inference
Frameworks & Programming	Python, PyTorch, TensorFlow, Keras, FastAPI, C, C++, Java
Platforms & Systems	Linux/Ubuntu, CUDA/GPU, HPC, Jupyter, Anaconda, Git/GitHub, VS Code, PyCharm, Cursor
Data & Applied AI	OCR, structured extraction, ETL, document intelligence, local models, Oracle, MySQL, Redis

PROFESSIONAL EXPERIENCE

Research Fellow

University Research Facility in Big Data Analytics (UBDA), The Hong Kong Polytechnic University, Hong Kong | Sep 2022 - Jun 2025

- Advised research teams on problem formulation, data preparation, model/framework selection, experimental design, GPU/HPC execution, debugging, performance analysis, and reproducible implementation.
- Designed, co-developed, and delivered hands-on training for PolyU researchers on LLMs, RAG/LangChain, GANs, image analytics, data analytics, CNN/RNN applications, medical imaging, and efficient image classification.
- Supported proposals and research plans with ML/DL methodology, data strategy, compute requirements, evaluation protocols, and implementation roadmaps.
- Conducted research on data-efficient and reconfigurable generative models, contributing to CVPR 2023, AAAI 2024, WACV 2025, IJCNN 2025, and ISDIA 2025.

Postdoctoral Fellow

Department of Computing & UBDA, The Hong Kong Polytechnic University, Hong Kong | Sep 2019 - Sep 2022

- Worked on RIF and ITF projects in food safety, big data, IoT, and non-targeted contaminant detection.
- Developed a smartphone-based, real-time, low-cost food-quality sensing prototype without external sensors; the demonstration received an A (Excellent) departmental evaluation.
- Researched GANs and unsupervised image-to-image translation, including model design, training, optimization, evaluation, and publication.

Senior Research Scholar / Research Scholar / Junior Research Fellow

Indian Institute of Technology (IIT) Roorkee, India | Jan 2015 - Aug 2019

- Developed SmartITS, a smartphone-sensor-cloud platform for seamless indoor-outdoor localization, crowd monitoring, trajectory tracking, and event detection.
- Built algorithms and proof-of-concept applications for spatio-temporal analytics, human-behaviour modelling, anomaly detection, and suspicious-activity identification.
- Published in IEEE Transactions on Mobile Computing, Journal of Network and Computer Applications, and IEEE conferences.

Earlier experience: Software Engineer, HCL Ltd. (Jul 2006 - Feb 2008); Assistant Professor roles at GLA University and Medi-Caps Institute.

SELECTED APPLIED AI & RESEARCH PROJECTS

- DocsMind - Document Intelligence: CPU-first pipeline for image/PDF → OCR → local language model → strict JSON → Excel; designed for privacy-aware document automation, field extraction, and modular document routing.
- Applied AI Product Prototyping: Practical, privacy-aware workflows combining data pipelines, model inference, analytics, automation, and deployable reporting components.
- BusinessAgent OS: Product direction for a privacy-first multi-agent operating layer that converts messages and documents into leads, tasks, quotes, follow-ups, and business reports.
- Data-Efficient Generative Models: Architectural reconfiguration, regenerative pruning, layer-specific pruning, and dynamic adaptation for GANs under limited-data and limited-compute settings.
- Smartphone Food Quality Sensing: Real-time visual sensing and deep-learning pipeline for low-cost food freshness assessment.
- SmartITS Crowd Intelligence: Seamless indoor-outdoor localization, crowd sensing, trajectory tracking, event detection, and mobile-cloud analytics.

SELECTED POLYU / UBDA TRAINING & WORKSHOPS

- Foundations and Applications of Large Language Models (2025): LLM architectures, pre-training, Transformers, RAG, LangChain, and demonstrations.
- Image Analytics in Practice (2024): Transfer learning, GAN training, super-resolution, inpainting, restoration, and generative applications.
- Transforming Images: GAN-Based Image-to-Image Translation (2024) and Practical Data Analytics I/II.
- Efficient and Accurate Image Classification; RNNs for Text Generation; Deep Learning for Medical Image Analytics (2022).
- Scalable Parallel Computing: Machine Learning with GPU (2021).
- CNN/RNN Applications on UBDA; Data-Driven Research and Proposal Writing (2020).

SELECTED PUBLICATIONS

1. IJCNN 2025 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, Y. Zhao, "Dynamic Generative Adaptation for Data-Efficient GAN."
2. WACV 2025 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "Data-Efficient Alignment in Medical Imaging via Reconfigurable Generative Networks."
3. AAAI 2024 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "RG-GAN: Dynamic Regenerative Pruning for Data-Efficient Generative Adversarial Networks."
4. CVPR 2023 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "Re-GAN: Data-Efficient GANs Training via Architectural Reconfiguration."
5. IEEE TIP 2022 - D. Saxena, T. Kulshrestha, J. Cao, S.-C. Cheung, "Multi-Constraint Adversarial Networks for Unsupervised Image-to-Image Translation."
6. IEEE TMC 2019 - T. Kulshrestha, D. Saxena, R. Niyogi, J. Cao, "Real-time Crowd Monitoring using Seamless Indoor-Outdoor Localization."
7. JNCA 2017 - T. Kulshrestha, D. Saxena, R. Niyogi, V. Raychoudhury, M. Misra, "SmartITS: Smartphone-based Identification and Tracking using Seamless Indoor-Outdoor Localization."

Full publication list: [Google Scholar](#)

EDUCATION, AWARDS & SERVICE

Ph.D., Computer Science & Engineering | Indian Institute of Technology (IIT) Roorkee, India | Jul 2015 - Aug 2019

- Best Paper Award - LS-GAN: Layer-Specific Pruning for Data-Efficient Generative Adversarial Networks, ISDIA 2025.
- Microsoft Research grant support, IIT Roorkee (2016); MHRD Scholarship for full-time Ph.D. studies.
- Reviewer: Nature, IEEE Access, Journal of Network and Computer Applications, and IEEE Sensors.

RESEARCH TRANSLATION & PROPOSAL SUPPORT

- Contributed ML/DL methodology, data/computing strategy, and implementation planning to the UBDA (Shenzhen) proposal.
- Contributed to the RGC Collaborative Research Fund equipment proposal "High Performance Deep Learning Clusters for Big Data Analytics" (2021/22).
- Through UBDA, supported researchers across PolyU departments in applying ML/DL to domain-specific projects, including method selection, experiment planning, implementation guidance, and proposal methodology.